B Chiranjeevi Adithya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.65
Expert PU College	Class XII (Karnataka School Examination and Asessment Board)	2024	97.33%
Aradhana School	Class X (ICSE)	2022	95.33%

ACHIEVEMENTS & PARTICIPATIONS

- **School Topper** in ICSE Class X Board Examination.
- **Expedia Group Campus Hackathon 2026** -- Participant

- Participated in the Basic Coding Round, applying data structures and algorithms to solve competitive programming challenges.

SKILLS

- **Programming Languages:** C, C++, Python, JavaScript , MySQL
- **Frontend:** HTML5, CSS3, Bootstrap, Tailwind CSS, React.js, Next.js, Redux Toolkit, Material UI
- **Backend:** Node.js, Express.js, REST APIs, Socket.io, WebSockets, JWT Authentication
- **Databases:** MongoDB, MySQL
- **Developer Tools:** Git, GitHub, Postman, Thunder Client, Linux CLI
- **Cloud & Deployment:** Render, Vercel, Cloudinary,Docker
- **Architecture & Concepts:** MVC Architecture, Client-Server Architecture, RESTful API Design, State Management, WebRTC, Real-Time Communication
- **DevOps & Cloud:** Docker, Kubernetes, CI/CD Pipelines, AWS, Cloudinary, Render,Vercel
- **Data Science & Visualization:** NumPy, Pandas, Matplotlib, Seaborn

PROJECTS

RAG Evaluation & Document Intelligence System | Python, LangChain, Qdrant, RAGAS, Sentence Transformers, LLMs | Github

- Built a **Retrieval-Augmented Generation (RAG)** pipeline for document question answering using adaptive chunking, Sentence Transformer embeddings, and **Qdrant vector retrieval**.
- Implemented and benchmarked **semantic retrieval and LLM generation**, evaluating responses using **RAGAS metrics including Faithfulness, Answer Relevancy, Context Precision, and Context Recall**.
- Achieved **1.00 Context Recall** on the evaluation dataset and analyzed retrieval quality using context precision (**0.39**) to identify areas for improving document retrieval.

Connecta | Github

Technologies: React.js, Node.js, Express.js, MongoDB, **WebRTC**, **Socket.io**, Material UI

- Developed a **real-time video conferencing platform** inspired by **Zoom** using **WebRTC** and **Socket.io** for **peer-to-peer communication**.
- Implemented **multi-user video/audio calling**, **screen sharing**, **in-meeting chat**, and **dynamic participant management**.
- Built **WebRTC signaling** using **SDP offer/answer negotiation**, **ICE candidate exchange**, and **STUN server integration**.
- Designed **RESTful APIs** with Express.js and MongoDB for **user authentication** and **meeting management**.
- Developed a **responsive React interface** using Material UI and Axios for seamless **frontend-backend communication**.

CodeForge | Github

Technologies: React.js, Node.js, Express.js, MongoDB, **JWT**, **AWS S3**, Socket.io

- Developed a **Git-based code repository platform** inspired by **GitHub** supporting **repository management**, **issue tracking**, and **collaborative development workflows**.
- Implemented **Git-inspired version control operations** including **init**, **add**, **commit**, **push**, **pull**, and **revert**.
- Built **secure RESTful APIs** using Express.js, MongoDB, **JWT authentication**, and **bcrypt-based password hashing**.
- Integrated **AWS S3** for scalable **repository and file storage** with a **modular backend architecture**.
- Developed an **interactive React interface** with Axios and **repository visualization** using **heat map components**.

WanderLust | Github

Technologies: Node.js, Express.js, MongoDB, **Passport.js**, **Cloudinary**, **Leaflet**

- Developed a **full-stack vacation rental platform** inspired by **Airbnb** for creating, browsing, updating, and managing **property listings**.
- Built **RESTful APIs** with complete **CRUD functionality** for listings and reviews using Express.js and MongoDB.
- Implemented **secure authentication**, **authorization**, and **session management** using **Passport.js** and **Express Session**.
- Integrated **Cloudinary**, **Multer**, and **Leaflet** for **image storage**, **file uploads**, and **interactive location mapping**.
- Improved application reliability through **Joi validation**, **centralized error handling**, and **flash messaging**.

RELEVANT COURSES

- **Computer Science**: Introduction to Programming, Data Structures & Applications, Artificial Intelligence.
- **Data Science & Analytics**: R and Python for Biologists, Big Data Biology and Biological Databases, Biostatistics.
- **Mathematics**: Probability ,Linear Algebra, Calculus ,Differential Equations.
- **Bioinformatics**: Next Generation Sequencing.

EXTRACURRICULAR

- National Service Scheme (NSS) Volunteer - Represented School in Cricket and Table Tennis